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AIR FORCE RESEARCH LABORATORY
(AFRL)

AIR FORCE RESEARCH LABORATORY
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Standardization

BIOSAFETY PROGRAM

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This instruction implements Department of Defense Manual (DoDM) 6055.18, *Safety Standards for Microbiological and Biomedical Laboratories*. It establishes policies and procedures and prescribes administrative and technical biosafety requirements for the implementation of the AFRL Institutional Biosafety Program. It applies to AFRL laboratories that use, handle, transport, transfer, store, or dispose of infectious agents and toxins (IAT), recombinant and synthetic nucleic acids, blood, and other potentially infectious materials (OPIM), and external organizations that have entered into an agreement with AFRL. All supplements to this instruction must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW Air Force Records Information Management System Records Disposition Schedule. Refer recommended changes and questions about this publication to the OPR using the DAF Form 847, *Recommendation for Change of Publication;* route Department of the Air Force (DAF) Form 847 from the field through the appropriate functional chain of command. Authorities to waive requirements in this publication are identified in the AFRL Delegation of Waiver Approval Authority of Tier Compliance Items Memorandum.

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1. Purpose.

1.1. Prescribes the biosafety requirements to use, handle, transport, transfer, store, or dispose of IAT, blood, OPIM, and recombinant and synthetic nucleic acids IAW DoDM 6055.18, National Institutes of Health (NIH) Guidelines, NIH and Centers for Disease Control and Prevention (CDC) Biosafety in Microbiological, Biomedical Laboratories (BMBL), the United States Government (USG) Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern (DURC) and Pathogens with Enhanced Pandemic Potential (PEPP) policy, and other applicable regulations and guidelines such as the Department of Transportation Hazardous Materials, and the International Air Transport Association regulations.

1.2. Implements the use of the AFRLMAN 60-102, *AFRL Biosafety Manual*, as the resource for information on institutional safety policies, procedures and laboratory practices for the possession, use, storage, disposal and transportation of IAT, blood, OPIM, and recombinant and synthetic nucleic acids. The AFRL Biosafety Manual is intended to serve as a primary resource for the development of laboratory and protocol-specific Laboratory Operating Protocols (LOPs).

1.3. Codifies the Institutional Biosafety Committee (IBC) as the reviewing and oversight body of the biological safety program and implementation of the program by the Biological Safety Officer (BSO).

1.4. Codifies the Institutional Review Entity (IRE). See **paragraph 3.1.8**.

2. Applicability.

2.1. All AFRL biomedical and biological research laboratories, environmental and public health laboratories, and animal laboratories rated animal biosafety level (ABSL)-2, and higher that use, handle, transport, transfer, store, or dispose of IAT, blood, OPIM, and recombinant and synthetic nucleic acids.

2.2. Oversight of the clinical laboratory is on an ad hoc basis and includes consult services for infectious waste and facility safety.

2.3. External organizations that have entered into an agreement with AFRL.

3. Roles and Responsibilities.

3.1. AFRL Commander/Director (AFRL/CC).

3.1.1. Serves as the Institutional Official (IO).

3.1.2. May delegate to the 711 HPW/CC, 711 HPW/DD, or other 711 HPW leadership, the respective duties of this section via Alternate Institutional Official (AIO) roles.

3.1.3. Appoints a Biological Safety Officer (BSO), who will also serve as the Institutional Contact for Dual Use Research (ICDUR), and an Alternate BSO to carry out the responsibilities of the BSO as required.

3.1.4. Establishes and supports a biological safety program staffed with a BSO and administrative, technical, and subject-matter expertise to support activities that involve IAT, blood, OPIM, recombinant and synthetic nucleic acids, and DURC and PEPP.

3.1.5. Ensures appropriate training in biosafety, risk management, personal protective equipment (PPE), engineering controls, primary and secondary containment, shipping and handling of IAT, blood, OPIM, recombinant and synthetic nucleic acids, and other topics are provided for IBC members, Principal Investigators (PIs), Laboratory Directors, and laboratory staff.

3.1.6. Ensures AFRL personnel comply with DoD and Department of the Air Force Occupational Health and Safety Program regulations and policies that include requirements to identify, assess, and control disease and injury risks to personnel from exposures.

3.1.7. Establishes governance, policies and procedures, and support for an IBC.

3.1.8. Appoints an IBC Chair and IBC Vice-Chair, voting and alternate voting members, and approve the selection of community members.

3.1.9. Reviews all committee proceedings.

3.1.10. When required, establishes governance, policies and procedures, and support for an IRE for the review of DURC and PEPP as defined by the USG DURC and PEPP policy. The IRE may be composed of IBC members but must function as a separate and independent entity or be established as a stand-alone committee as necessary based on the breadth of expertise required and the volume of work involving DURC and PEPP.

3.1.11. Appoints an IRE Chair and IRE Vice-Chair, and at least four additional voting members.

3.1.12. Reviews all proceedings of the entity.

3.1.13. Ensures implementation of risk mitigation plan for and oversight of DURC and PEPP activities.

3.1.14. Provides education and training on DURC and PEPP for individuals conducting life sciences research that falls within the scope of USG DURC and PEPP policy. Maintain records of training for three years.

3.1.15. Ensure that within 30 calendar days, non-compliance with USG DURC and PEPP policy and mitigation measures undertaken to prevent recurrences of similar non-compliance, is reported to the Program Officer of the Federal funding agency or to the NIH if the research is non-Federally funded.

3.2. **Biological Safety Officer (BSO).**

3.2.1. Serves as biosafety subject matter expert (SME) and provides and supports risk assessments, risk management, biosafety controls, biological safety program management, LOPs, biosafety training, inspections, mishap notification, investigation and reporting, and emergency planning and response.

3.2.2. Must be federal employees (military or civilian) and meet the education, experience, training, and credentialing requirements for biosafety officers as specified in DoDM 6055.18.

3.2.3. Responsible for the implementation, maintenance, conduct of and compliance with the AFRL Biological Safety Program. Duties include:

3.2.3.1. Assist PIs, Laboratory Directors, Supervisors, Staff, the IBC, and IRE regarding development and implementation of policies and procedures to reduce the risks of work with biohazardous materials with consideration given to having minimal interference with the conduct of research and teaching.

3.2.3.2. Review of research and laboratory activities involving IAT, blood, OPIM, and recombinant and synthetic nucleic acids.

3.2.3.3. Prepare and distribute a Biosafety Manual, which will be reviewed at least annually and revised as necessary.

3.2.3.4. Assist in the development of plans and provide advice regarding emergency management, handling of biological mishaps and incidents, and personnel exposure.

3.2.3.5. Ensure the implementation of infectious waste disposal policies and procedures to comply with state, federal, and DoD regulations.

3.2.3.6. Conduct biosafety inspections. Whenever possible, inspections will be coordinated with other inspecting entities.

3.2.3.7. Review laboratory-specific biosafety plans and LOPs prior to implementation.

3.2.3.8. Investigate biological mishaps and incidents involving IAT, blood, OPIM, recombinant and synthetic nucleic acids, and DURC and PEPP. Report details and violations, if found, to IO and the appropriate office. Within the BSO's discretion, plans and LOPs will be reviewed and recommendations provided to prevent subsequent biological mishaps and incidents.

3.2.3.9. Provide organizations with standardized and up-to-date training materials for biosafety, biohazard communication, and bloodborne pathogen (BBP) training; and ensure PIs and Laboratory Directors have current information for laboratory-specific and agent-specific training for laboratory personnel.

3.2.3.10. Provide assistance to PIs and Laboratory Directors in performing laboratory risk assessments, as requested.

3.2.3.11. Serve as a voting member on the IBC. BSO will also act as liaison and SME to other committees such as the Institutional Animal Care and Use Committee (IACUC), Institutional Review Board (IRB), Radiation Safety, and others as applicable.

3.2.3.12. When requested, will assist the Competent Medical Authority with annual review of occupational illness and injury reports to determine if revision of exposure prevention strategies is indicated.

3.2.3.13. The BSO will coordinate with other offices according to written agreements and any other external organizations (e.g., 88 Operational Medical Readiness Squadron (OMRS), Occupational Health (SGX), AFRL, Safety (RQY), IACUC, etc.) to address biosafety practices and considerations with respect to:

3.2.3.13.1. Occupational-health risk assessment and mitigation requirements.

3.2.3.13.2. Microbiological and biomedical laboratory design and commissioning.

3.2.3.13.3. Access control.

3.2.3.13.4. Chemical hygiene, including toxins.

3.2.3.13.5. Facility, utilities and equipment maintenance.

3.2.3.13.6. Integrated pest management.

3.2.3.13.7. Operations with radioactive material.

3.2.3.13.8. Biospecimens collected from human subject's research participants.

3.2.3.13.9. Work with animals.

3.2.3.13.10. Work with invertebrate vectors and hosts.

3.2.3.13.11. Emergency planning and response.

3.2.3.13.12. Biological mishap investigation and reporting.

3.2.3.14. The BSO will serve as the ICDUR.

3.2.3.14.1. The ICDUR assists with the development of risk-mitigation plans IAW USG DURC and PEPP policy and serves as the liaison between the institution and the relevant program officer at the Federal funding agencies, or between the institution and the NIH for non-federally funded dual-use research.

3.2.3.14.2. The ICDUR will provide an annual formal assurance of institutional compliance to federal agencies that fund DURC and PEPP.

3.3. **The Institutional Biosafety Committee (IBC) Chair.**

3.3.1. Presides over IBC meetings and approves meeting minutes.

3.3.2. Ensure non-compliance, biological mishaps, laboratory inspection findings, and corrective actions are documented in IBC minutes.

3.3.3. Ensures the IBC member composition complies with the requirements of the NIH Guidelines.

3.3.4. Provides Biosafety training for IBC members as outlined in the Collaborative Institutional Training Initiative Program. (See references)

3.3.5. Ensures conflicts of interest within the committee are identified at the onset of each meeting and managed accordingly.

3.3.6. Pre-reviews IBC protocols in addition to assigning two IBC members for pre-review.

3.3.7. Notifies PIs of the results of the IBC's review.

3.3.8. Notifies the IO if proposed research may be categorized as DURC and PEPP.

3.4. **The Institutional Review Entity (IRE) Chair.**

3.4.1. Must have the capability to assess the risks associated with the release or sharing of information, knowledge and technologies from DURC and PEPP.

3.4.2. Preside over IRE meetings and approve meeting minutes.

3.4.3. Ensures institutional compliance with USG DURC and PEPP policy.

3.4.4. Serves as point of contact (POC) for DURC and PEPP reviews and risk-mitigation plans.

3.4.5. Ensures annual reports are received by the PI and reviewed by the IRE.

3.4.6. Ensures conflicts of interest within the entity are identified at the onset of each meeting and managed accordingly.

3.4.7. Reports any violations of the USG policy to the IO and ICDUR.

3.5. **Institutional Biosafety Committee (IBC) and Institutional Review Entity (IRE) Vice-Chairs.**

3.5.1. Have the same authority as the Chairs, when chair is not present.

3.5.2. Serve as alternates to the respective Chairs.

3.5.3. Execute the responsibilities of the Chairs as required.

3.6. **Supervisory Personnel.**

3.6.1. Supervisory personnel who provide daily oversight of laboratory staff (e.g., PIs, Laboratory Directors, Supervisors, etc.) are responsible for day-to-day compliance with AFRL biosafety requirements and the implementation of local, state, and federal regulations that apply to laboratory activities.

3.6.2. PIs, Laboratory Directors and Supervisors.

3.6.2.1. Possess detailed knowledge about infectious agents to the extent that he/she can ensure AFRL bio-labs are compliant with biosafety regulations/practices. Implement the AFRL Biosafety manual, (and laboratory specific instruction manuals).

3.6.2.2. Ensure the safe operation of laboratories; proper use, maintenance, and care of equipment; and safe handling of biohazardous materials and waste. Having physical presence in the laboratory to maintain compliance and able to respond in a timely manner to any onsite incidents.

3.6.2.3. Ensure personnel are enrolled in an occupational health program, when required, IAW DAFI 48-145, *Occupational and Environmental Health Program*. Comply with Workplace Supervisor requirements in DAFI 48-145.

3.6.2.4. Assess the risks of laboratory activities involving biohazardous materials and investigate all safety aspects of planned work. If potentially high risk-group pathogen is isolated from environmental samples or other materials, the BSO should be notified immediately.

3.6.2.5. Inform all individuals in the laboratory of potential hazards associated with the work and document communication of the hazards to the individual. Laboratory supervisor enforces AFRL-IBC policies that control access to the laboratories. Restricting access to non-authorized people.

3.6.2.6. Provide initial and annual refresher training for laboratory personnel in safe work practices and maintain records of personnel training.

3.6.2.7. Notify the BSO prior to the initiation of new activities to ensure a documented risk assessment and laboratory inspection is conducted.

3.6.2.8. Submit protocols using the electronic IBC Review System (IBCR) available on the Human Performance Wing Support (HPWS) website.

3.6.2.9. Review and document review of each LOP under them every 12 months or when revisions are necessary. Prior to initial implementation and upon annual review, the BSO and other qualified safety personnel (e.g., system safety, occupational and environmental health, etc.) should review and concur with the LOP. Guidance for LOPs is provided in the AFRLMAN 60-102 and DoDM 6055.18. Simultaneously, the PI will review the associated AFRL Form 5, Lab Safety Permit, and coordinate with the Site Safety Office to ensure the permit accurately reflects current laboratory hazards and configurations. Guidance for LOPs is provided in the AFRLMAN 60-102 and DoDM 6055.18. Guidance for AFRL Form 5 is provided in DAFI 91-202_AFRL Supplement.

3.6.2.10. Conduct and document monthly inspection of laboratories (supervisor may designate an individual to conduct monthly inspections).

3.6.2.11. Inform the BSO of notification of a scheduled inspection by an external agency (e.g. CDC, Environmental Protection Agency, Department of Transportation, etc.).

3.6.2.12. Inform the BSO of any biological mishaps and incidents, and non-compliance issues within three business days of receiving a report. Potentially life-threatening events or laboratory-acquired infections must be reported immediately.

3.6.2.13. Be knowledgeable about and comply with all institutional and Federal policies and requirements for oversight of DURC and PEPP.

3.6.2.14. Contact the IRE Chair if his or her research involves one or more of the agents or toxins listed in **paragraph 6.2.1** of this AFRLI.

3.6.2.15. Work with the IRE to develop risk mitigation measures.

3.6.2.16. Conduct DURC and PEPP in accordance with the provisions in the risk mitigation plan.

3.6.2.17. Ensure laboratory personnel conducting life sciences research that falls within the scope of USG DURC and PEPP policy have received education and training on DURC and PEPP.

3.7. **Laboratory Personnel.**

3.7.1. Responsible for their own safety and for personnel around them. Maintain a positive attitude about safety.

3.7.2. Comply with all DAFI 48-145, *Occupational and Environmental Health Program* requirements, laboratory-specific, agent-specific LOPs, and federal, state and institutional policies. Reference AFRLMAN 60-102 for more information.

3.7.3. Complete required laboratory safety and biosafety training prior to engaging in activities with IAT, blood, OPIM, or recombinant and synthetic nucleic acids.

3.7.4. Report any hazardous or potentially unsafe conditions to the laboratory supervisor or unit-level safety officer.

3.7.5. Inform the supervisor or BSO of any biological mishaps and incidents or non-compliance issues within one business day. Potentially life-threatening events must be reported immediately.

3.8. AFRL Personnel can notify the Biosafety Office and call for a review of laboratories or actions believed to be not in compliance with appropriate safety guidelines, of protocol violations, or of possible work being conducted without an approved protocol.

4. Laboratory Inspections.

4.1. Laboratory inspections will be conducted by the BSO (and/or BSO designees).

4.2. The BSO may conduct initial inspections for PIs requesting IBC approval for new projects.

4.3. All AFRL active laboratories covered by this instruction will be inspected at least semi-annually (excluding the clinical Biosafety Level (BSL)-2 laboratories).

4.4. Inspections include review of LOPs, personnel training records, safety equipment calibration certificates, spill response plans, and other documents as requested.

4.5. Periodic and unannounced laboratory inspections may be conducted, as necessary, to investigate biological mishaps and incidents, reports of non-compliance, and to evaluate the effectiveness of the AFRL Biosafety Program.

4.6. BSL-2 and BSL-3 checklists provided on the Biosafety SharePoint site will be used to conduct inspections.

4.7. Inspection reports will be provided to the PI, laboratory safety POC, and supervisor within 10 business days.

4.7.1. Copies provided to the safety office if requested or deemed necessary by the BSO.

4.8. Deficiencies or procedures that create a potentially life-threatening situation will be immediately referred to the supervisor. The operation will be stopped and corrective actions immediately implemented.

5. Institutional Biosafety Committee Procedures.

5.1. The IBC serves as an expert safety advisory, review, and approval board for laboratory activities that involve the use of IAT, blood, OPIM, and recombinant and synthetic nucleic acids, and performs the following duties IAW DoDM 6055.18, and applicable NIH guidelines, the BMBL, and other federal guidelines and regulations.

5.1.1. IAW DODM 6055.18, at a minimum, the committee must:

5.1.1.1. Review proposed laboratory activities and facility modifications.

5.1.1.2. Assist in performing biological risk assessments.

5.1.1.3. Discuss mishaps and near misses.

5.1.1.4. Review results of compliance inspections.

5.1.2. In addition, the IBC:

5.1.2.1. Recommend biocontainment levels.

5.1.2.2. Determines if proposed research activities and experimental outcomes meet the definition of DURC and PEPP.

5.1.2.3. Assists with the development and implementation of emergency plans covering emergency management, biological mishaps and incidents, and personnel contamination.

5.1.2.4. Proposes institutional policy and implementation plans when required (e.g. following biological mishaps or incidents, due to changes in regulations and guidelines, etc.).

5.1.2.5. Conducts periodic self-evaluations of their functions and operations including this AFRLI, guidelines, policies, procedures, checklists, etc.

5.2. **Membership.**

5.2.1. The committee will consist of qualified occupational and environmental health personnel, equipment and facility maintenance and safety personnel, and members of the workforce.

5.2.2. Membership will include appointed and ex-officio (non-voting) members.

5.2.2.1. Appointed Members

5.2.2.1.1. In addition to an appointed Chair, the IBC must have no fewer than five voting members and will consist of:

5.2.2.1.1.1. BSO.

5.2.2.1.1.2. PIs or Clinical Laboratory Director.

5.2.2.1.1.3. Two members unaffiliated with AFRL.

5.2.2.1.1.4. A representative of the laboratory staff.

5.2.2.1.1.5. Animal, plant, and human subjects research experts.

5.2.2.1.2. IBC voting members are appointed by the IO and may include DoD service members (active, reserve, national guard), and DoD civilians. Government Contractors are exempted from being IBC voting members.

5.2.2.1.3. Alternate voting members may be appointed by the IO to serve in the absence of any regular voting member or when business before the committee represents a conflict of interest for a regular voting member.

5.2.2.2. Ex-officio members. Includes members that are SMEs from research and clinical laboratories, occupational health, industrial hygiene, facility maintenance, safety, and security.

5.2.3. Ad hoc SMEs may be used, when necessary, to provide consultation on matters that are unique or outside the normal expertise of the IBC.

5.3. **Protocol Review.**

5.3.1. The laboratory activities in this section must be reviewed and approved by the IBC. Submission to the IBC will be accomplished using IBCR. Reviews include: (i) an assessment of the containment levels required for the proposed research; and (ii) assessment of the facilities, procedures, practices, and training and expertise of personnel.

5.3.2. IBC Chair Review/Approval. Activities categorized as BSL-1 containment, NIH-exempt experiments, and BSL-2 involving the use of blood or OPIM only, may be reviewed by the BSO and IBC Chair, and approved by the IBC Chair. The IBC Chair may choose to forward any protocol for full committee review. PIs/POCs should be available to present the research to the IBC upon request by IBC Chair. IBC members will be notified at the next scheduled meeting of protocols approved by the IBC Chair.

5.3.3. Convened IBC Review/Approval.

5.3.3.1. Activities categorized as BSL-2 containment involving IAT in Risk Group 2 or higher agents, involving recombinant and synthetic nucleic acids in Risk Group 2 and higher, any research that involves the introduction of recombinant or synthetic nucleic acids or IAT into human subjects or whole animals, and all research activities requiring BSL-3 containment must be reviewed and approved by the IBC.

5.3.3.2. For research involving human subjects, PIs must comply with DoDI 3216.02_AFI 40-402, Protection of Human Subjects and Adherence to Ethical Standards in DoD/AF-Supported Research, and AFRLI 40-402, Protection of Human Subjects in Research.

5.3.3.3. For research involving whole animals, PIs must comply with DoDI 3216.01, Use of Animals in DoD Conducted and Supported Research and Training, and DAFPD 40-4, Protection of Animal and Human Subjects.

5.3.3.4. Research activities involving the possession, use or transfer of inactivated IAT derived from organisms that are classified as Biological Select Agents (BSAT, 42 CFR Part 73), certain Risk Group 2, and all Risk Group 3 or Risk Group 4 microorganisms. This includes microorganisms that have been rendered “inactivated” or otherwise non-viable (e.g., by chemical treatment, irradiation, etc.) and genomic material or proteins derived from microorganisms.

5.3.4. Laboratories accredited by the College of American Pathologists that conduct non-research activities are not required to submit an IBC protocol or receive IBC approval prior to the initiation of testing activities.

5.3.5. **Amendments.**

5.3.5.1. Include changes that occur to the research processes, laboratory facilities, addition of IAT, and updates to personnel list. PIs must receive IBC approval prior to implementing the amendment(s).

5.3.5.2. Must be submitted by modifying the protocol in IBCR.

5.3.5.3. Amendments that do not increase the risk may be approved by the IBC Chair. Those in which risk is increased will be reviewed and approved by the IBC.

5.3.6. Re-approvals. PIs must submit a request for re-approval via IBCR 30 days prior to expiration. Re-approval should include, as applicable, amendments, updated training, personnel list, LOPs, and IRB approvals, etc. Protocols that have lapsed in expiration must cease research activities. It is the PI's responsibility to track the expiration dates of their protocols and supporting documents. The PI will receive an automated notification 60 days prior to expiration for the protocol only.

5.3.7. Closures. The PI must contact the Biosafety Office to request protocol closure. The Biosafety Office will update the protocol status in IBCR to "closed."

5.4. **Meetings.**

5.4.1. The IBC will meet regularly at least monthly. Ad hoc meetings will be held when deemed necessary by the IBC Chair.

5.4.2. A quorum consisting of a simple majority (half of the voting membership plus one) is required to approve protocols.

5.4.3. An IBC review packet will be provided prior to the meeting and will contain:

5.4.3.1. An agenda that includes: business items and a list of protocols that did not meet a convened board and were approved by the IBC Chair.

5.4.3.2. Prior meeting minutes for review and approval, if not previously approved electronically.

5.4.3.3. Other documents as required (e.g. training documents, supporting documentation).

5.4.4. The PI, or an associate investigator, will attend the IBC meeting to present their application to the committee and be available to answer committee questions directly. The PI must be excused prior to board deliberation and voting.

5.4.5. Any member of the committee may call for a vote on any issue(s) being reviewed, discussed, or decided by the IBC for the record.

5.4.6. Votes will be carried electronically by a simple majority as long as quorum is maintained. If a regular voting member is unavailable, an alternate will be polled in his or her place. Voting and any abstentions will be indicated verbally or by a show of hands and recorded.

5.4.7. The IBC administrator will take meeting minutes. Minutes will be provided to the IBC chair for preliminary review, and subsequently to IBC members who were in attendance for the convened meeting. Once approved, the IBC Chair will sign the minutes and provide them to the IO for review. Minutes will be maintained for three years.

5.4.8. **IBC Determinations.**

5.4.8.1. Approved without modification or restriction. The IBC may approve laboratory activities as submitted.

5.4.8.2. Approved pending modifications. Laboratory activities may be approved by the committee with modifications pending. The IBC may vote to grant the IBC Chair the authority to approve modifications without bringing them back to the full committee or entity before final approval is granted. Activities cannot be initiated until final approval is issued in writing.

5.4.8.3. Conditional Approval/Approval with Restriction(s). The IBC may approve laboratory activities with conditions or restrictions. The condition or restriction must be explicit in the minutes and in the PI notification. A justification for the condition or restriction must also be presented along with instructions to the PI on the satisfaction of the condition(s) and if/how/when the restriction may be removed.

5.4.8.4. Deferred. The IBC may defer the approval of laboratory activities pending receipt of modifications, additional information, and/or clarifications. Any deferred laboratory activities will be included in the agenda for the next regularly scheduled IBC meeting and shall be reconsidered upon receipt of requested information by a specified deadline.

5.4.8.5. Disapproved. The reasons for disapproval of laboratory activities must be explicit in the notification to the PI. The PI must also be provided with any instructions for resubmission that will meet the criteria for approval.

5.4.9. PI Notification. On behalf of the IBC, the IBC Chair or IBC Program Manager will provide email notification regarding actions taken or decisions made by the committee.

5.4.10. Approval will be for three years, unless otherwise determined by the IBC.

5.4.11. IBC, IRE, and Biosafety Officer determinations may be appealed for reconsideration. Reconsideration requests shall be submitted to the Chair of the IBC, IRE or Biosafety Officer. If after reconsideration, further appeal is desired, second reconsideration request shall be submitted to the AIO. If further relief is desired after AIO determination, appeal may be made to the AFRL IO. In any case, the AIO and IO may disapprove an item that was previously approved or seek modifications to obtain approval, however no entity may approve an item that the IBC, IRE, or Biosafety Officer disapproved."

6. Institutional Review Entity Procedures.

6.1. The IRE is a committee required by USG DURC and PEPP policy to assist with DURC and PEPP assessment, determination, and risk mitigation plan development.

6.2. According to the USG DURC and PEPP policy, an IRE DURC and PEPP assessment is required for research that uses one of the agents listed in 6.2.1. below, and produces, aims to produce, or can reasonably be anticipated to produce one or more of the effects listed in 6.2.2. below. AFRL is not registered to possess, use, or transfer any Select Agents with the exception of permissible amounts of toxins.

6.2.1. **Agents and Toxins.**

6.2.1.1. All Biological Select Agents and Toxins (BSAT), regulated by the Federal Select Agent Program (FSAP).

6.2.1.2. All risk group 4 pathogens.

6.2.1.3. A subset of risk group 3 pathogens.

6.2.1.4. Pathogens that are not assigned a risk group by the National Institutes of Health and recommended to be handled at Biosafety Level 3, (BSL3), and Biosafety Level 4 (BSL4).

6.2.2. Research Experimental Outcomes.

6.2.2.1. Increase transmissibility of a pathogen within or between host species;

6.2.2.2. Increase the virulence of a pathogen or convey virulence to a non-pathogen;

6.2.2.3. Increase the toxicity of a known toxin or produce a novel toxin;

6.2.2.4. Increase the stability of a pathogen or toxin in the environment, or increase the ability to disseminate a pathogen or toxin;

6.2.2.5. Alter the host range or tropism of the agent or toxin;

6.2.2.6. Decrease the ability for a human or veterinary pathogen or toxin to be detected using standard diagnostic or analytical methods;

6.2.2.7. Increase resistance of a pathogen or toxin to clinical and/or veterinary prophylactic or therapeutic interventions;

6.2.2.8. Generate, use, reconstitute, or transfer an eradicated or extinct PPP or a previously identified PEPP;

6.2.2.9. Alter a human or veterinary pathogen or toxin to disrupt the effectiveness of pre-existing immunity, via immunization or natural infection, against the pathogen or toxin; or

6.2.2.10. Enhance the susceptibility of a host population to a pathogen or toxin.

6.3. The PI is responsible for notifying the Biosafety Office as soon as:

6.3.1. PI's research involves any of the agents listed in **paragraph 6.2.1**.

6.3.2. PI's research with one or more of the above agents also produces or can be reasonably anticipated to produce one or more of the effects listed in **paragraph 6.2.2**.

6.3.3. PI's research that meets the criteria in **paragraph 6.2** may meet the definition of DURC and PEPP.

6.4. After notification by the Biosafety Office, the IRE Chair will:

6.4.1. Request the PI complete an AFRL Request for IRE Determination of DURC and PEPP and submit to the IRE Chair. The Request for IRE Determination of DURC and PEPP is located on the Biosafety SharePoint site: **https://usaf.dps.mil/sites/20938/1/SitePages/Biosurety.aspx**. (From the home page navigate to the Biosafety Templates folder, or to the DURC and PEPP link on the left side of the home page. If the link is inaccessible, contact the Biosafety Office).

6.4.2. Upon receipt of the Request for IRE Determination of DURC and PEPP, the IRE Chair will schedule IRE meeting(s) to:

6.4.2.1. Verify research involves the agents listed in 6.2.1.

6.4.2.2. Review the PI's assessment.

6.4.2.3. Make a final determination on the applicability of the list of experimental effects in 6.2.2.

6.4.2.4. Conduct a risk assessment to determine whether the research meets the definition of DURC and PEPP. The risk assessment should consider that knowledge, information, products, or technologies resulting from the research could be used in a manner that results in harm or threatens society. IRE should provide DURC and PEPP determination to the ICDUR and the ICDUR must notify the USG funding agency within 30 calendar days of the IRE determination. The PI may appeal to the IRE Chair regarding the IRE's determination, and a subsequent meeting will be held to reassess the determination. If the PI and IRE continue to disagree, the IO will make the determination.

6.4.2.5. Draft a risk mitigation plan.

6.4.2.6. Work with the USG funding institution to complete the draft mitigation plan within 90 calendar days of the IRE's determination that the research is DURC and PEPP.

6.5. **The IRE must:**

6.5.1. Consist of at least five members from the IBC who have received DURC and PEPP training.

6.5.2. Be sufficiently empowered to ensure compliance with the institutions' dual use research policies.

6.5.3. Have sufficient experience within the membership to assess dual-use potential conducted at or sponsored by the institution.

6.5.4. Have knowledge of dual-use issues, concerns, and related institutional and Federal policies, and understand risk-assessment and risk-management considerations.

6.5.5. Make its procedures for reviewing life-sciences research for dual-use potential accessible to the public IAW institutional policies and procedures for the release of information; however, IRE minutes will not be subject to public disclosure.

6.5.6. Maintain records of institutional DURC and PEPP review and risk-mitigation plans for three years.

6.5.7. Review all active risk-mitigation plans at least annually.

6.5.8. Have separate meetings and minutes from the IBC.

6.6. Subject matter experts will include POCs from Security and 88 ABW/PA (Public Affairs). Other SMEs will be identified as relevant to specific activities identified as life sciences DURC and PEPP. At least one SME must have experience and expertise in research involving recombinant or synthetic nucleic acid molecule technology.

6.7. Once named, the IRE will conduct a self-evaluation of their functions and operations IAW USG policy, DoD policy and AFRL policies, procedures, etc. This review will be an agenda item for the initial meeting and will be recorded in the minutes along with any proposed action steps.

7. Conflicts of Interest.

7.1. Members of the IBC and IRE may not be involved in the review or approval of a laboratory or project in which he/she has been or expects to be engaged or has a direct financial interest.

7.2. Members are required to identify potential conflicts of interest at the onset of every IBC and IRE meeting.

7.3. If a member is uncertain of their conflict, they may recuse themselves at any time or table the matter pending clarification.

7.4. A member is not required to provide the reason for recusal.

8. Agreements, Contract Activities, and Permits.

8.1. Contracts will be written to ensure that contract personnel comply with this AFRLI and other applicable regulations. The Biosafety Office will assist Contracting Officers (COs) to include appropriate biosafety provisions in Professional Work Statements upon request.

8.2. External organizations that conduct biological research and require biosafety support will have to enter into a formal agreement with AFRL and this agreement will be coordinated through the AFRL/IR Biosafety Office.

8.3. Material transfer agreements (MTAs) involving the transfer or purchase and shipment of biological materials from vendors and collaborators will be coordinated through the AFRL/IR Biosafety Office.

8.3.1. The unit Technology Transfer Office and the CO will coordinate through their Judge Advocate or Commander, as applicable.

8.4. Import and Export Permits must be coordinated through the Biosafety Office.

DOUGLAS P. WICKERT,
Brigadier General, USAF
Commander, Air Force Research Laboratory

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 33-322, *Records Management and Information Governance Program,* 28 July 2021

AFRLI 40-402, *Protection of Human Subjects in Research*, 17 March 2023

DAFI 48-145, *Occupational and Environmental Health Program,* 22 September 2022

DoDI 3216.01, *Use of Animals in DoD Conducted and Supported Research and Training,*

20 March 2019

DoDI 3216.02_AFI 40-402, *Protection of Human Subjects and Adherence to Ethical Standards in Air Force Supported Research*, 29 June 2022

DoDM 6055.18, *Safety Standards for Microbiological and Biomedical Laboratories,*

11 August 2020

CFR 29 Part 1910.1030, *Bloodborne Pathogens,* **https://www.ecfr.gov/cgi-bin/retrieveECFR?gp=1&SID=b9126e9fba23e3e7933354a1d2630d72&ty=HTML&h=L&n=9y1.0.1.5.58&r=PART**

49CFR Part 171-185. *Hazardous materials Regulations*.

Defense Transportation Regulations 4500.9-R-Part II, Chapter 204, *Hazardous Material*,

16 May 2024

International Air Transport Association, *Dangerous Goods Regulations,* Current Edition

Memorandum for Cindy Coe, Occupational Safety and Health Administration (OSHA) Regional Administrator from Richard Fairfax, Directorate of Enforcement Programs, "*Response to Interpretation Request for Application of the Bloodborne Pathogens Standard to Veterinary Clinics,"* 15 October 2002 (**https://www.osha.gov/laws-regs/standardinterpretations/2002-10-15**)

National Institutes of Health, *NIH Guidelines for Research Involving Recombinant and Synthetic Nucleic Acid Molecules (NIH Guidelines),* April 2024

National Institutes of Health and Centers for Disease Control and Prevention (CDC), *Biosafety in Microbiological and Biomedical Laboratories (BMBL),* Sixth Edition, June 2020

National Institutes of Health*, Tools for the Identification, Assessment, Management, and Responsible Communication of Dual Use Research of Concern: A Companion Guide to the United States Government Policies for Oversight of Life Sciences Dual Use Research of Concern,* September 2014

United States Government *Policy for Oversight of Dual Use Research of Concern and Pathogens with Enhanced Pandemic Potential, May 2024*

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication under Adopted Forms*

Abbreviations and Acronyms

ABSL—Animal Biosafety Level

ABW—Air Base Wing

AFI—Air Force Instruction

AIO—Alternate Institutional Officer

AVMA—American Veterinary Medical Association

BBP—Bloodborne Pathogen

BMBL—Biosafety in Microbiological and Biomedical Laboratories

BSAT—Biosafety and Select Agents and Toxins

BSL—Biosafety Level

BSO—Biological Safety Officer

CDC—Centers for Disease Control and Prevention

CFR—Code of Federal Regulations

CO—Contracting Officer

DoD—Department of Defense

DoDI—Department of Defense Instruction

DoDM—Department of Defense Manual

DURC—Dual Use Research of Concern

HBV—Hepatitis B Virus

HIV—Human Immunodeficiency Virus

HPW—Human Performance Wing

HPWS—Human Performance Wing Support

IACUC—Institutional Animal Care and Use Committee

IAT—Infectious agents and toxins

IAW—In Accordance With

IBC—Institutional Biosafety Committee

IBCR—Institutional Biosafety Committee Review

ICDUR—Institutional Contact for Dual Use Research

IO—Institutional Official

IRB—Institutional Review Board

IRE—Institutional Review Entity

LOP—Laboratory Operating Procedure

MTAs—Material Transfer Agreements

NIH—National Institutes of Health

OI—Operating Instruction

OPIM—Other Potentially Infectious Material

OPR—Office of Primary Responsibility

OSHA—Occupational Safety and Health Administration

PAM—Pamphlet

PEPP—Pathogens with Enhanced Pandemic Potential

PI—Principal Investigator

PPE—Personal Protective Equipment

POC—Point of Contact

RG—Risk Group

SME—Subject Matter Expert

TD—Technical Directorate

USG—United States Government

Terms

BIOLOGICAL MISHAP—An event in which the failure of laboratory facilities, equipment, or procedures appropriate to the level of potential pathogenicity or toxicity of a given etiologic agent (organism or toxin) may allow the unintentional, potential exposure of humans or the laboratory environment to that agent.

BIOMEDICAL LABORATORY—A laboratory that utilizes biological science for medical research, development, test, and evaluation.

BIOSAFETY LEVEL—Provide guidelines for the safe use of infectious organisms and are based on the Risk Group of the organism, laboratory practices and techniques, engineering controls and building containment features. The levels of containment range from the lowest BSL-1 to the highest at level-4 (BSL-4).

BLOOD—According to CFR 29 Part 1910.1030, *Bloodborne Pathogens*, human blood, human blood components, and products made from blood. The American Veterinary Medical Association (AVMA) recommends voluntary compliance with OSHA's BBP standard in order to best protect employees working in veterinary or other animal settings. The referenced JAVMA article identifies exposure to animal blood as a potential risk for the transmission of diseases (JAVMA, Vol. 204, No. 3). Blood, for the purposes of this OI, therefore, also includes animal blood, blood components and products made from animal blood.

BLOODBORNE PATHOGENS—According to CFR 29 § 1910.1030, *Bloodborne Pathogens*, pathogenic microorganisms that are present in human blood and can cause disease in humans. These pathogens include, but are not limited to, hepatitis B virus (HBV) and human immunodeficiency virus (HIV).

DUAL USE RESEARCH—Research conducted for legitimate purposes that generates knowledge, information, technologies, and/or products that can be utilized for benevolent or harmful purposes.

DUAL USE RESEARCH OF CONCERN—Life sciences research that, based on current understanding, can be reasonably anticipated to provide knowledge, information, products, or technologies that could be directly misapplied to pose a significant threat with broad potential consequences to public health and safety, agricultural crops and other plants, animals, the environment, materiel, or national security.

Institutional Contact for Dual Use Research (ICDUR)—The designee who serves as the institutional point of contact for questions regarding compliance with and implementation of the requirements of the USG DURC and PEPP policy. The ICDUR assists with the development of risk-mitigation plans IAW USG DURC and PEPP policy and serves as the liaison between the institution and the relevant program officer at the Federal funding agencies, or between the institution and the NIH for non-federally funded dual-use research.

INFECTIOUS AGENTS AND TOXINS (IAT)—Fungi, virus, bacteria, prions, rickettsia, parasites, or a viable microorganism or its toxin or a prion that lacks nucleic acids and that causes or may cause disease, and any material of biological origin that poses a degree of hazard similar to those organisms. Term IAT includes Biological Select Agents and Toxins (BSAT). BSAT are select microorganisms and biological toxins listed in 42 CFR § 73, 9 CFR § 121, and 7 CFR § 331 that have the potential to pose a severe threat to public, animal, or plan health or to animal or plant products. The possession, use and transfer of BSAT is regulated by the CDC and U.S. Department of Agriculture.

INSTITUTION—An organization (institute, agency, center, etc.), a contract organization such as a school of medicine, or a research institute that conducts research, development, test, evaluation, or sampling and analysis with IAT.

INSTITUTIONAL BIOSAFETY COMMITTEE—Committee appointed by the Institutional Official to develop and review policies and procedures regarding the use of IAT, blood, OPIM, and recombinant and synthetic nucleic acids.

INSTITUTIONAL OFFICIAL—The commanding officer or institute director of an activity conducting research, development, test, evaluation, or sampling and analysis with IAT. The vice commanding officer or institute vice director is the Alternate Institutional Official and will act on behalf of the Institutional Official as required.

INSTITUTIONAL REVIEW ENTITY—Committee appointed by the Institutional Official reviews research with agents listed in DURC and PEPP and DoD DURC and PEPP Implementation Guidance to determine if it is DURC and/or PEPP and when research is determined to be DURC or PEPP, draft a risk mitigation plan for identified DURC and PEPP activities.

LABORATORY—An individual room or rooms within a facility that provides space in which work with IAT can be performed. It contains all of the appropriate engineering features and equipment required at a given BSL to protect personnel working in it and the environment external to the facility.

MICROBIOLOGY—The science and study of microorganisms, including protozoans, algae, fungi, bacteria, viruses, and prions.

MOLECULAR BIOLOGY—A branch of biology that deals with the molecular basis of biological activity; chiefly concerns itself with understanding the interactions between the various systems of a cell, including the interactions between the different types of deoxyribonucleic acid, ribonucleic acid and protein biosynthesis as well as learning how these interactions are regulated.

OTHER POTENTIALLY INFECTIOUS MATERIAL (OPIM)—Includes: 1) The following human body fluids: semen, vaginal secretions, cerebrospinal fluid, synovial fluid, pleural fluid, pericardial fluid, peritoneal fluid, amniotic fluid, saliva in dental procedures, all body fluids visibly contaminated with blood and all body fluids in situations where it is difficult or impossible to differentiate between body fluids; 2) Any unfixed tissue or organ (other than intact skin) from a human (living or dead); and 3) HIV containing cell or tissue cultures, organ cultures, and HIV- or HBV- containing culture medium or other solutions; and blood, organs, or other tissues from experimental animals infected with HIV or HBV; 4) based on AVMA recommendations, OPIM also includes the animal fluids listed in 1) and unfixed animal tissue or organs (other than intact skin); 5) any patient or animal specimen that contains or has a greater than minimal likelihood to contain infectious agents.

PATHOGENS WITH ENHANCED PANDEMIC POTENTIAL RESEARCH (PEPP)—Research on biological agents and toxins, which when enhanced have the potential to pose risk to the public health, agriculture, food security, economic security, or National security.

RECOMBINANT AND SYNTHETIC NUCLEIC ACIDS—See below.

a)—Molecules that are constructed by joining nucleic acid molecules and b) that can replicate in a living cell, i.e., recombinant nucleic acids;

b)—Nucleic acid molecules that are chemically or by other means synthesized or amplified, including those that are chemically or otherwise modified but can base pair with naturally occurring nucleic acid molecules, i.e., synthetic nucleic acids, or

c)—Molecules that result from the replication of those described in a) or b) above.

RISK ASSESSMENT—An assessment of the probability that harm, injury, or disease will occur. In the context of the microbiological and biomedical laboratories, risk assessment focuses primarily on the prevention of laboratory-associated infections. Risk assessment is used to assign the BSLs (facilities, equipment, and practices) that reduce to an absolute minimum worker and environment risk of exposure to an agent.

RISK GROUP—Classification of agents according to their relative pathogenicity for healthy adult humans by the following criteria: (1) Risk Group 1 (RG1) agents are not associated with disease in healthy adult humans. (2) Risk Group 2 (RG2) agents are associated with human disease which is rarely serious and for which preventive or therapeutic interventions are often available. (3) Risk Group 3 (RG3) agents are associated with serious or lethal human disease for which preventive or therapeutic interventions may be available. (4) Risk Group 4 (RG4) agents are likely to cause serious or lethal human disease for which preventive or therapeutic interventions are not usually available. (NIH Guidelines definition).

TOXIN—Toxic material of biologic origin that has been isolated from the parent organism; the toxic material of plants, animals, or microorganisms.